

Exclusive J/ψ Electroproduction at CLAS12: Monte Carlo Tuning with the Forward Tagger and NN Fast Simulation

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Abstract

Tuning of the DIFFRAD Monte Carlo generator for exclusive J/ψ electroproduction $ep \rightarrow e'p J/\psi$, $J/\psi \rightarrow e^+e^-$, with the scattered electron in the CLAS12 Forward Tagger (quasi-photon production, $Q^2 \lesssim 0.1 \text{ GeV}^2$). The model uses a dipole t -dependence. The combined CLAS12 RGA sample of 68 J/ψ candidates (52 at 10.6 GeV, 16 at 10.2 GeV) is acceptance-corrected with neural-network fast simulation and fitted by MC reweighting.

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1 Cross-Section Model

$$\frac{d\sigma_T}{dt} = \alpha_1 \left(1 - \frac{W_{\text{th}}^2}{W^2}\right)^{\alpha_2} W^{\alpha_3} \left(1 + \frac{Q^2}{m_\psi^2}\right)^{-\nu_T} \cdot \frac{3(m_g^2 - t_{\min})^3}{(m_g^2 - t)^4} \quad (1)$$

$$\frac{d\sigma_L}{dt} = c_R \frac{Q^2}{m_\psi^2} \frac{d\sigma_T}{dt}, \quad W_{\text{th}} = M_N + m_\psi = 4.035 \text{ GeV}. \quad (2)$$

Here $t, t_{\min} < 0$ and the dipole factor is normalized: $\int_{-\infty}^{t_{\min}} 3(m_g^2 - t_{\min})^3 / (m_g^2 - t)^4 dt = 1$, so the t -integrated cross section is carried entirely by the W and Q^2 factors. Near threshold $|t|_{\min}$ is large (2.7 GeV² at W_{th} , 1.0 GeV² at $W = 4.5$), which makes the correct $t_{\min}$ treatment essential.

In Forward Tagger kinematics ($Q^2 \in [0.015, 0.1]$ GeV²) the factors $(1 + Q^2/m_\psi^2)^{-\nu_T}$ and $c_R Q^2/m_\psi^2$ vary by less than 1%: the model has no usable Q^2 lever arm, and ν_T, c_R, α_3 are kept fixed. The free parameters are α_2 (threshold behaviour) and m_g^2 (dipole t -slope).

2 Monte Carlo Generator

`diffrad_jpsi_dipole.f90` (DIFFRAD family, one particle = one generator). All model parameters ($\alpha_1, \alpha_2, \alpha_3, \nu_T, m_g^2, c_R$) are read from the input file. The accept-reject proposal uses the same dipole shape, so the sampling efficiency tracks m_g^2 automatically.

Two bugs were found and fixed during this work:

- **Dipole normalization sign:** the code evaluated $(m_g^2 - |t_{\min}|)^3$ instead of $(m_g^2 + |t_{\min}|)^3$. Since $|t|_{\min} > m_g^2$ below $W \approx 4.23$ GeV, the cross section went negative and threshold events were silently rejected.
- **32-bit overflow:** the attempt limit `maxtry = 100000*nev` overflowed `INTEGER*4` for $n_{\text{ev}} \geq 2\text{M}$, terminating generation instantly. Counters are now `integer*8` (also fixed in the ϕ generator).

Samples: 5M events at 10.6 GeV, 2M at 10.2 GeV. Generation ranges: $Q^2 \in [0.01, 0.12]$, $W \in [4.05, 4.55]$, $|t| < 6$ GeV², $\theta_{e'} \in [0^\circ, 7.5^\circ]$.

3 Data

Combined RGA J/ψ candidate sample (Mariana), 68 events (52 at 10.6 GeV, 16 at 10.2 GeV), four detected particles: e' in the Forward Tagger ($\theta = 2.7\text{--}6.0^\circ$), decay e^+e^- and recoil proton in the Forward Detector. $M(e^+e^-) \in [2.93, 3.17]$ GeV — the sample is pre-selected, the yield is obtained by counting (no mass fit). 64 events pass the analysis cuts ($Q^2 \in [0.015, 0.1]$, $W \in [4.15, 4.5]$, $|t| < 3.0$).

4 Fast Monte Carlo and Acceptance

NN models (acceptance + smearing) trained per particle and per run configuration: e^- (FT, no vertex measurement, v_z fixed at -3 cm), decay e^-, e^+, p . The 10.6 GeV MC is evaluated against all 8 run configurations (F18 in/out, S18 in/out at their beam currents); each event's acceptance weight is the fraction of configurations that accept it. The 10.2

GeV MC uses S19in. The two beam energies are then combined so that the *accepted* MC matches the data composition 52:16.

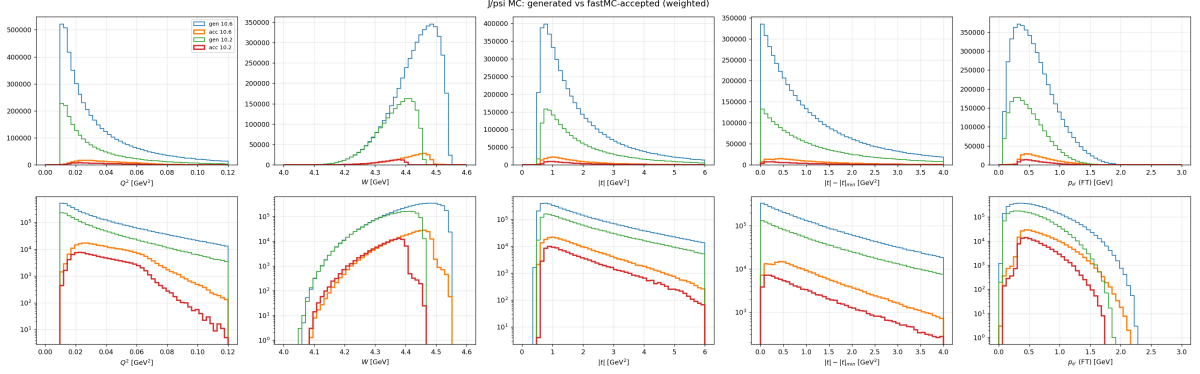


Figure 1: Generated vs fastMC-accepted distributions for both beam energies.

5 Tuning Iterations

Fit: acceptance-corrected counts in $3 Q^2 + 3 W + 4 |t|$ bins (10 points), model predictions by event-by-event reweighting $\sigma_{\text{new}}/\sigma_{\text{gen}}$, free parameters α_2 and m_g^2 , **differential_evolution** minimization.

Table 1: Tuning iterations. Starting values $\alpha_2 = 1.0$, $m_g^2 = 1.6$ (Bhawani/lAger).

Iter	α_2^{gen}	$m_g^{2,\text{gen}}$	α_2^{fit}	$m_g^{2,\text{fit}}$	χ^2/ndf
1	1.000	1.600	3.719	3.144	24.1/8
2	3.719	3.144	4.048	3.152	17.9/8
3	4.105	3.106	4.122	3.112	17.3/8

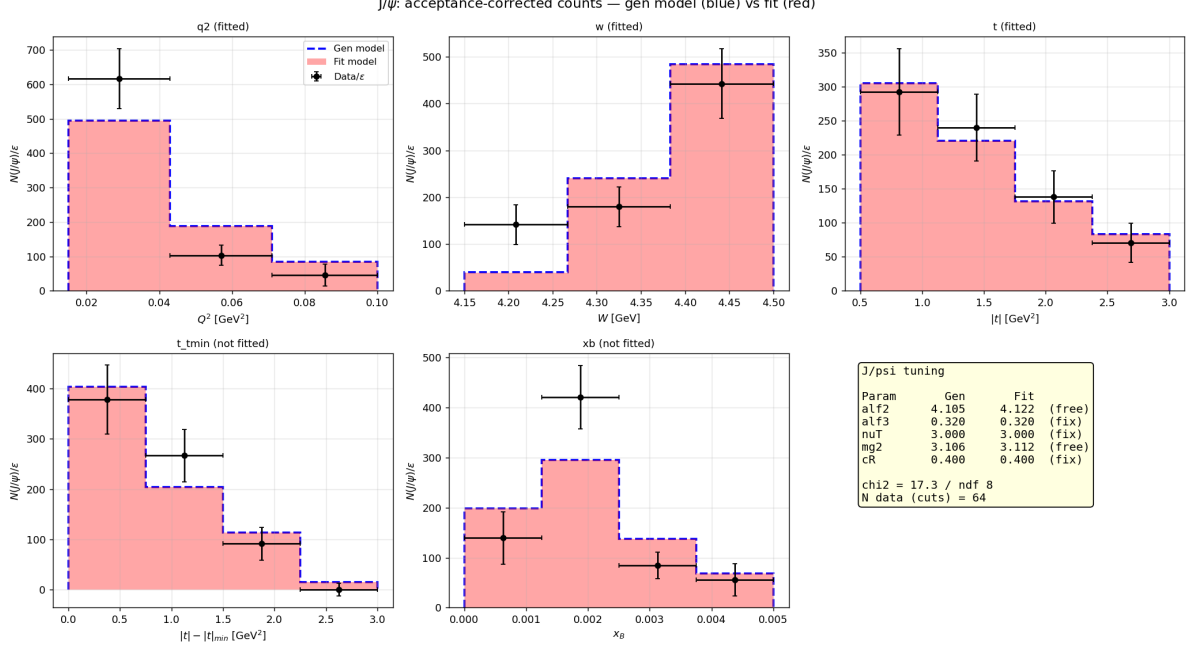


Figure 2: Acceptance-corrected counts vs generation model (blue) and fitted model (red). Q^2 , W , $|t|$ are fitted; $|t| - |t|_{min}$ and x_B are cross-checks. The first Q^2 bin carries the dominant χ^2 contribution; the model Q^2 shape is flux-driven and not tunable in FT kinematics.

6 Cross-Section Shapes in the (W, Q^2) Analysis Bins

Flux-folded $\gamma^* p \rightarrow J/\psi p$ shapes in the three correlated (W, Q^2) bins of the CLAS12 analysis (Table 16 of the analysis note). Absolute normalization requires the integrated luminosity and is not attempted here.

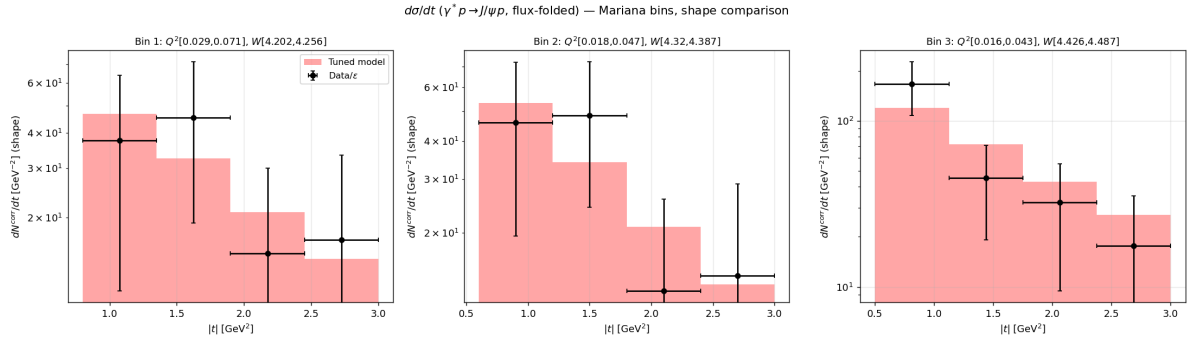


Figure 3: $d\sigma/dt$ shape in the three (W, Q^2) bins: acceptance-corrected data (points) vs tuned model (red).

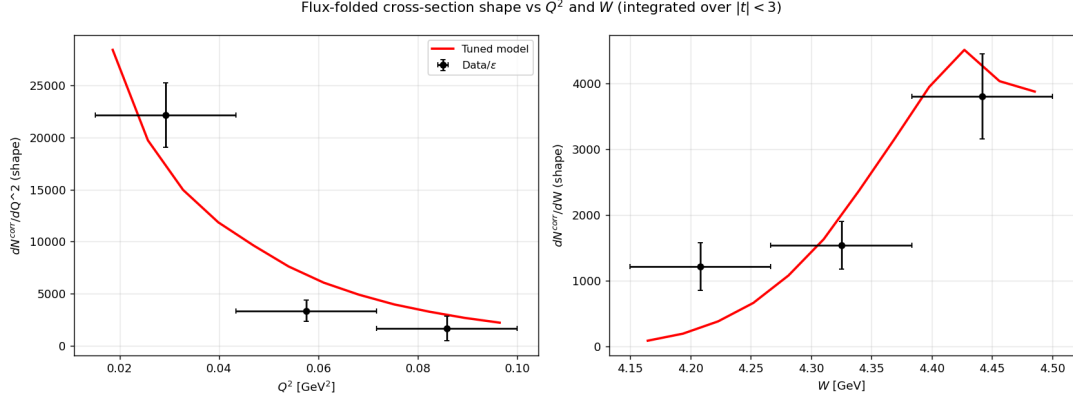


Figure 4: Cross-section shape vs Q^2 and W , integrated over $|t| < 3$ GeV².

7 Acceptance Comparison with the CLAS12 Analysis

Table 2: Our NN-fastMC mixture acceptance vs the efficiencies of the CLAS12 analysis (Table 17): η_1 event-by-event, η_{2a}/η_{2b} bin-averaged with two methods.

Bin	Q^2 [GeV ²]	W [GeV]	ε (this work)	η_1	η_{2a}	η_{2b}	N_{data}
1	[0.029, 0.071]	[4.202, 4.256]	0.1016	0.1463	0.1184	0.1406	7
2	[0.018, 0.047]	[4.320, 4.387]	0.1142	0.1101	0.1096	0.1386	11
3	[0.016, 0.043]	[4.426, 4.487]	0.0869	0.1090	0.1127	0.1313	15

8 Summary

The tuned J/ψ generator parameters are $\alpha_2 = 4.12$, $m_g^2 = 3.11$ GeV² ($\alpha_3 = 0.32$, $\nu_T = 3.0$, $c_R = 0.4$ fixed), $\chi^2 = 17.3$ for 8 ndf. The data prefer a much steeper threshold rise and a flatter t -dependence than the starting (photoproduction-inspired) parametrization. With 68 events the two-parameter description is adequate; the remaining tension is confined to the lowest Q^2 bin, where the model shape is fixed by the virtual photon flux.